# *CS/EE/IDS 166: Computational Cameras*

*Course Syllabus – Spring 2026*

California Institute of Technology

**Course Instructor**

Katherine (Katie) Bouman
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Office Hours: Wednesday 1:30-3:00 pm in Annenberg 346
(except 4/8, 4/22, 5/20, 6/3 – by appointment to replace these dates)

**Teaching Assistants**

Brandon Zhao
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Office Hours: Thursday 3-4:30 PT in Annenberg 104

Brady Bhalla
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Office Hours: Thursday 7-8 PT in Annenberg 121

Zachary Huang
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Office Hours: Monday 4-5:30 PT in Annenberg 104

**Course Description**

Computational cameras overcome the limitations of traditional cameras, by moving part of the image formation process from hardware to software. In this course, we will study this emerging multi-disciplinary field at the intersection of signal processing, applied optics, computer graphics, and vision.

**Course Website and Online Resources**

*Lectures (Tues/Thurs 1:00-2:30 PT): Annenberg 104*

*Canvas:* https://caltech.instructure.com/courses/9727

*Piazza:* https://caltech.instructure.com/courses/9727/external_tools/9660

*Gradescope:* https://caltech.instructure.com/courses/9727/external_tools/7048
Entry Code: V4X5YN

**Assessment Rubric**

The 5 homework assignments are worth 65% of your grade, the final project is worth 30%, and participation (through Piazza, etc) is worth 5%. Late homework assignments will be penalized by 20% for every day they are late; however, the first 5 late days (total, across all 5 assignments) will be forgiven. Late final projects cannot be accepted, as the registrar sets a hard deadline for grade reports.

**Participation**

There will be a participation grade for the class that can be completed as a combination of attending lectures and/or being active on Piazza and/or office hours.

**Collaboration Policy**

The HW you hand in must be your own and not copied from others. You are encouraged to work on the problems with others and to seek additional help if you find that useful, but the write-up and code must be your own. You may not look at any other student's homework (write-up or code). Also, you may not consult any prepared solutions for the problems, whether they are this years or from previous years, or from Caltech or external sources. Additionally, you may not use solutions from course material at other schools or post your answers online (this includes *e.g.* public GitHub repositories). You cannot use ChatGPT for help on any homework problem or for help with your final project report, except in rare cases when granted permission by the instructor.

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|---|---|
| Basic discussion of the problems | Yes |
| Look at communal materials while writing up solutions | No |
| Look at other's non-communal work (i.e., writeups, code) | No |
| Turn in a set with more than one name on it<br>(does not include naming collaborators in basic discussion) | No |
| Use of LLM and/or AI agent for help in assignments | No on homework<br>May be used with instructor permission in the final project |

**Academic Integrity**

Caltech's Honor Code: "No member of the Caltech community shall take unfair advantage of any other member of the Caltech community."

Understanding and Avoiding Plagiarism: Plagiarism is the appropriation of another person's ideas, processes, results, or words without giving appropriate credit, and it violates the honor code in a fundamental way. You can find more information at: http://writing.caltech.edu/resources/plagiarism.

All instances of plagiarism or other academic misconduct will be referred to the Board of Control for review.

**Students with Documented Disabilities**

Students who may need an academic accommodation based on the impact of a disability must initiate the request with Caltech Accessibility Services for Students (CASS). Professional staff will evaluate the request with required documentation, recommend reasonable accommodations, and prepare an Accommodation Letter for faculty dated in the current quarter in which the request is being made. Students should contact CASS as soon as possible, since timely notice is needed to coordinate accommodations. http://cass.caltech.edu/. Undergraduate students should contact Dr. Lesley Nye, Associate Dean of Undergraduate Students and graduate students should contact Dr. Kate McAnulty, Associate Dean of Graduate Studies.

**Course Schedule**

| *Week* | *Date* | *Lecture Topic* | *Homework* |
|---|---|---|---|
| *1* | *March 31* | *Syllabus & Computational Cameras* | |
| *1* | *April 2* | *Color* | |
| *2* | *April 7* | *Pinholes & Lenses* | |
| *2* | *April 9* | *Digital Photography* | *HW 1 released* |
| *3* | *April 14* | *Noise & Filtering* | |
| *3* | *April 16* | *Exposure & High Dynamic Range Photography* | *HW 1 due*<br>*HW 2 released* |
| *4* | *April 21* | *Pyramids & Blending* | |
| *4* | *April 23* | *NO CLASS* | |
| *5* | *April 28* | *Light Fields & Focal Stacks* | *HW 2 due*<br>*HW 3 released* |
| *5* | *April 30* | *Signal Processing & Linear Spatial Filters* | |
| *6* | *May 5* | *Linear Inverse Problems & Deconvolution* | *HW 3 due*<br>*HW 4 released* |
| *6* | *May 7* | *Coded Aperture & Lensless Cameras* | |

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| *7* | *May 12* | *Compressed Sensing & Single Pixel Cameras* | *HW 4 & final project proposal and literature review due*<br>*HW 5 released* |
| *7* | *May 14* | *Radio Interferometry & Black Hole Imaging* | |
| *8* | *May 19* | *Deep Learning for Inverse Problems (guest lecture by Berthy Feng)* | *HW 5 due* |
| *8* | *May 21* | *Tomography & Single-view 3D Reconstruction*<br>*(guest lecture by Brandon Zhao)* | |
| *9* | *May 26* | *Phase Retrieval & Applications*<br>*(guest lecture by Jason Wang and Rodrigo Ferrer-Chavez)* | |
| *9* | *May 28* | *Diffraction Imaging*<br>*(guest lecture by Hosea Nelson and Dmitry Eremin)* | *Interim project report due* |
| *10* | *June 2* | *Finals Week – No Lecture* | |
| *10* | *June 4* | *Finals Week – No Lecture* | *Final projects and presentations due* |